MedTok – Week 7 Progress Report

Validation on linked MIMIC-IV multimodal data

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1. Goal of this week

You asked me to use the MIMIC-IV multimodal data from Rafi to test and validate my method. Until now I tested on three separate public datasets (MedMNIST, PhysioNet 2012, Cataract-101), and my clinical text was synthetic. MIMIC lets me check three open questions: (1) do the tokenizers work on real hospital data, (2) does the unified protocol hold when one patient has several real modalities at once, and (3) is real clinical text handled losslessly?

2. Data

Table 1. The linked demo subset (p10–p19). Join key: subject_id.

Item	Value
Patients (train / val / test, split by patient, stratified by label)	600 (419 / 89 / 92)
Chest X-ray images (MIMIC-CXR-JPG, all views)	1083
Radiology reports (real free text)	600
12-lead ECGs (MIMIC-IV-ECG), 7 days before the X-ray	848
Lab tests kept / median results per patient	40 / 29.5
Died within 1 year of the X-ray (label)	90 (15.0 %)
Median age / male fraction	65 / 0.50

Timeline of one patient: t = 0 is seven days before the X-ray; labs and ECGs keep their real times; the X-ray and report are at t = 7 days. Nothing after the X-ray is used. The data is processed on our server only; this report contains aggregate numbers only.

3. What I did

- **Method:** each modality goes through an existing tokenizer: labs → event tokens, ECG → window VQ tokens, X-ray → image VQ tokens (K = 1,024), report → byte-level BPE sub-words. All parts of a patient are written into one protocol v0.2 sequence in time order.
- **Addition 1 – RoutedCodec:** ECG and labs are both 'time series' but need different tokenizers. A small router picks the codec from the data source and stores the source in the sequence, so decoding needs no extra information.
- **Addition 2 – exact long gaps:** time-gap tokens were exact only up to 48 h (longer gaps used rounded buckets). Long gaps are now written as a

chain of exact ≤ 48 h tokens. Gaps ≤ 48 h give the same tokens as before, so all earlier PhysioNet results are unchanged.

- **Images:** the MedMNIST image tokenizer was used unchanged (zero-shot), and the same design was trained on MIMIC X-rays (28 px and 64 px). All 62 unit and integration tests pass.

4. Results

Table 2. Unified protocol on real linked records.

Check (all 600 patients)	Result
Sequences passing structural validation	600 / 600
Save → load of the token shard identical	100.0 %
Report text exact after the round-trip	100.0 %
Lab observation pattern exact after the round-trip	100.0 %
Sequence length mean / max (tokens)	1,128 / 4,394
Compressed size per patient	2.3 KiB

Table 3. Chest X-ray reconstruction. Higher PSNR/SSIM = closer to the original.

Image tokenizer (test, 163 images)	Size	PSNR dB	SSIM	Active codes	Tokens / image
MedMNIST, zero-shot	28 px	25.39	0.935	498 / 1024	56
MIMIC-trained	28 px	29.18	0.966	1018 / 1024	56
MIMIC-trained	64 px	31.60	0.949	1024 / 1024	272

Table 4. ECG and lab tests.

ECG / labs	Result
ECG: tokens per 10-s ECG / compression vs 100 Hz 16-bit	251 / 76×
ECG: mean lead correlation (test)	0.728 ± 0.100
ECG: PRDN % (lower = better)	83.6 ± 15.6
Labs (6 h grid): tokens per patient / pattern exact	139 / 100.0 %
Labs: test patients with a gap > 48 h (affected by the fix)	6.5 %

Table 5. Real radiology reports.

Text tokenizer (92 test reports)	Vocabulary	Exact round-trip	Tokens / report
BPE trained on synthetic notes	972	100.0 %	379 ± 132
BPE trained on	1,024	100.0 %	166 ± 71

MIMIC reports			
BPE trained on MIMIC reports	4,096	100.0 %	124 ± 44

Table 6. Death within 1 year (90 of 600). 5x repeated stratified 5-fold CV, logistic regression (L2, C=0.1, balanced class weights) on standardised log1p token counts. Chance: AUROC 0.5, AUPRC 0.15.

Input tokens	AUROC	AUPRC
image	0.633 ± 0.073	0.285 ± 0.069
ecg	0.578 ± 0.056	0.209 ± 0.044
labs	0.694 ± 0.060	0.346 ± 0.079
text	0.670 ± 0.053	0.283 ± 0.066
all (image+ecg+labs+text)	0.778 ± 0.047	0.383 ± 0.091
demographics	0.755 ± 0.070	0.377 ± 0.109
all + demographics	0.781 ± 0.047	0.386 ± 0.091

5. Key findings

- **The protocol works on real data:** all 600 patient sequences pass validation and come back exactly after saving and loading. This is the main claim of the thesis, now shown on real linked records.
- **Real data found a bug:** 6.5 % of test patients have a lab gap over 48 h; before the fix their lab times were decoded wrongly. Synthetic tests and 48-h PhysioNet stays could not show this.
- **Images:** the MedMNIST codebook reads real X-rays but uses only part of its codes and blurs more; the same design trained on MIMIC uses almost all codes and is clearly sharper. The design generalises; the codebook should be trained on in-domain images.
- **ECG:** it works with no code changes and high compression, but keeps only the rough waveform shape. This is the weakest part.
- **Real text:** lossless for every tokenizer; the synthetic-trained vocabulary needs more than twice as many tokens per report as a vocabulary trained on real reports.
- **Tokens are informative:** every modality alone is above chance and all four together are best. Age and sex alone are already strong, so I claim 'informative and combinable tokens', not a mortality model.

6. Limitations

- **Size:** 600 patients from one hospital system: a validation of the method, not a clinical model.
- **Resolution:** X-rays are downsampled to 28 / 64 px, so the image metrics measure tokenizer fidelity, not diagnostic quality.
- **Label:** mortality is a convenience label; cross-validation spread is wide.

- **Data use:** all processing stays on our server; models trained on MIMIC are not shared.

7. Next steps

- **Improve the ECG tokenizer:** more codes per window or a convolutional encoder.
- **Shared codebook:** train one image codebook on MedMNIST + MIMIC and test it on both.
- **Thesis:** write the MIMIC validation chapter of the thesis.

8. Questions for you (please write your answer under each question)

Q1. Is this MIMIC validation enough for the thesis, or should I use more MIMIC data (for example, more patients)?

Your answer: _____
_____
_____

Q2. Is death within 1 year a good downstream task, or do you prefer another label (for example, findings from the report, or in-hospital mortality)?

Your answer: _____
_____
_____

Q3. Should the MIMIC results go in the main text of the thesis or in the appendix?

Your answer: _____
_____
_____

Q4. The ECG tokenizer is the weakest part. Should I improve it now, or report it as a limitation?

Your answer: _____
_____
_____

Q5. Should I try one shared image codebook for MedMNIST + MIMIC, or keep separate codebooks?

Your answer: _____
_____
_____

Q6. Is it acceptable to show only aggregate numbers (no MIMIC images) in the thesis and the defence slides?

Your answer: _____
_____
_____

Q7. Do you need the code or the results in another format?

Your answer: _____
_____
_____

Source of all numbers: reports/mimic/MIMIC_RESULTS.md and mimic_metrics.json, generated by python -m medical_tokenizer.mimic_validate. Code: github.com/REDOANUZZAMAN/medtok (docs/mimic_validation.md).